

Marcus Oates

marcus@moates.com.au | 0428 211 020 | [LinkedIn](#) | Portfolio: [moates.com.au](#)

Profile

- First employee and backend engineer at Voxworks, a Voice AI startup, where I led a ground-up restructure of the voice agent backend, its core product, and scaled call concurrency by 400%.
- Creator of Gym Junkie, a fitness tracking app now live on the Google Play Store and Apple App Store.
- Automation software engineer at Downer, building automated microservices with ML/AI, managing Azure resources and spearheading IaC adoption (GitHub Actions).
- Bachelor of Engineering (First Class Honours) / Computer Science (AI), UNSW: WAM 84.
- Short-term consultancy internship at Incat Crowther in naval engineering and architecture (2023).

Experience

Senior Software Engineer: Voxworks (Voice AI Startup)

Jan 2026 – Present

- As the startup's first employee, took operational control of the voice agent's backend architecture and led a ground-up restructure of the codebase, the company's core product, owning its reliability and ongoing development in production.
- Rebuilt the core conversation logic while maintaining interoperability with the existing live system, so clients continued to be served without interruption.
- Extended reliable call handling from ~2-minute calls to consistently sustaining 10–20 minute conversations that stay on script and complete their objectives.
- Built automated testing for the voice system, from unit and integration tests to concurrent automated call runs, using AI agents to analyse call dynamics and debug conversational issues across a large sample space.
- Established automated CI/CD deployment pipelines with real-time observability and monitoring in Grafana, giving instant visibility into call performance and system health.
- Converted the voice agent to a chat agent while maintaining the core integrations, primarily used for client testing.
- Architected the agent's conversational dynamics for a 10% improvement in average response latency, prioritising accuracy so the agent does not stray from its call objectives.
- Developed agentic AI development workflows and custom MCP tooling that dramatically accelerated development time and enabled rapid prototyping, under heavy human oversight with disciplined review to keep code quality high.
- Increased call concurrency by 400% through system optimisations, managing concurrency and cost-effectiveness across the pipeline to balance throughput against infrastructure spend.

Automation Software Engineer: Downer Group

May 2023 – Dec 2025

- Automated data extraction from spreadsheets using Azure Function Apps, stored in a database with a Power BI connection, saving ~\$175k a year per business-unit implementation and eliminating manual transcription errors.
- Created a custom internal Azure dashboard for monitoring project resources across subscriptions, with resource actions and deployment monitoring.
- Reduced operating costs of Azure resources by ~60% by scheduling startup and shutdown events through a centralised tagging policy enacted via the dashboard.
- Automated Konect API queries for critical field work, notifying the requisite authorities to consistently meet SLAs and avoid total penalties of ~\$450k.
- Enhanced security through daily storage account key rotation and SAS generation via Azure Key Vault, reducing 80–100 points of failure to a single centralised point.
- Extracted text with OCR from CAD drawing PDFs and generated project report spreadsheets, a time saving of ~95%.
- Wrote project setup shell scripts that deploy environments and resources in Azure, linked to custom GitHub branch environments with rules and protections to enable consistent IaC deployments using Bicep.
- Implemented bespoke deep learning and computer vision solutions for real-time artefact recognition with ~94% accuracy.

Engineering Intern: Incat Crowther

Feb 2023 – May 2023

- Produced accurate technical drawings with AutoCAD LT, focusing on double-hull commercial vessels of 30–120 ft.
- Created and modified technical frame and construction drawings for 3 vessels, quickly learning how boats are constructed and outfitted for service.

- Complied with relevant classing authority engineering standards so vessels were safe to operate under varying oceanic conditions.
- Produced precise cut-part, engine and rudder modification, and machinery arrangement drawings for clients within a dynamic environment.

Education

Bachelor of Engineering (Honours) / Computer Science (AI), UNSW Sydney

2019 – 2023

- Honours WAM: 84 (13 High Distinctions, 7 Distinctions).
- Notable marks: MTRN2500 Computing for Mechatronics = 98, COMP3311 Database Systems = 97, COMP3121 Algorithms and Programming Techniques = 96, MMAN3200 Linear Systems and Control = 95.

Higher School Certificate, St Augustine's College Sydney

2018

- ATAR 97.30. Aggregate Cup Recipient (2018), First XI Football team (2018), Stan Arneil Memorial Award Recipient (2017).

Projects

Gym Junkie

- Free, powerful workout tool and data tracker. React Native frontend (Expo) with a FastAPI (Python) backend on Postgres and Redis.
- Live on the Google Play Store and Apple App Store (Android and iOS).
- Lightweight frontend with a focus on increased functionality over existing paid apps.
- Enables users to track their workout progression over time with graph displays.
- Leaderboards, exercise history and muscle maps help users understand their progression.

Other

- Finska: lightweight game tracker for the popular game.
- DownerHelper: PyPI package, a collection of wrappers to eliminate code debt across projects.
- Cell Tracking: segmented cells in a sequence and tracked their positions over time.
- Postgres Deploy: deploys and mirrors DB schemas from a JSON configuration.

Technical Skills

AI / LLMs: Conversational voice agents; LLM fine-tuning and prompt engineering across frontier and open-source models; OpenAI, Anthropic and Groq; speech-to-text (Deepgram) and text-to-speech (Cartesia); RAG with vector search; custom MCP servers; LLM evaluation pipelines and automated testing at scale.

Azure: Function Apps (standard & durable, Python), Container Registry & Instances, Managed Identity, Key Vault, Storage Accounts (+ queues), Event Grid, Postgres; IaC with Bicep and GitHub workflows, az CLI; SharePoint and Logic Apps automation; VM deployment and monitoring; containerised Streamlit apps for ML/AI demos.

AWS: ECS, Secrets Manager, Lambda (zip/containerised), NLB, ElastiCache (Redis), RDS Postgres, CloudFormation, aws CLI.

Python: ML/AI with TensorFlow, OpenCV, scikit-learn, pandas, Pillow; FastAPI for RESTful interfaces; psycopg2 and asyncpg.

Telephony / Networking: SIP, RTP and WebRTC signalling; FreeSWITCH; real-time voice media handling.

Web / Mobile: React (TS), React Native (TS), Express, Jotai.

Other: SQL, PL/pgSQL, Bicep, Docker, C/C++, Shell, HTML, CSS, Java, Cloudflare; SolidWorks, AutoCAD LT and Rhino for complex naval mechanical designs, schematics and visual demonstrations.

Extracurricular

Triathlon, mountain biking, bouldering and gym.

Referees

Andrew D'Ambrosio · Jared Buncome · Mark Volz · Andrew Tuite

Contact details available upon request.